FINAL YEAR PROJECT

AEOS

Autonomous Engineering Orchestration System

From Chatbots to Autonomous Engineering Teams

The Elevator Pitch

> *AEOS is an operating system for engineering infrastructure. Instead of only answering questions like a chatbot, AEOS runs a team of specialised AI agents that actually take action — fixing broken builds, upgrading cloud servers, patching security holes, and creating pull requests — while always asking for approval before making dangerous changes.*

The Problem We Are Solving

Today's Reality: The "AI Assistant" Limitation

What Current AI (ChatGPT, Copilot) Does	What They Cannot Do
Answer questions about code	Access your actual cloud infrastructure
Generate code snippets	Modify running Kubernetes clusters
Explain error messages	Fix a broken CI/CD pipeline automatically
Suggest improvements	Create pull requests with validated fixes
Write documentation	Remember past incidents and learn from them

The Gap We Identified

A developer notices that an EC2 instance's CPU has hit 95% utilisation.

Current AI

- "You should consider upgrading your instance type."
- The developer is left to ask: Okay, but how? And I still have to do it manually.

AEOS

- "I've analysed CloudWatch metrics. Your t2.micro is overloaded. I recommend upgrading to t3.medium. This will cause about 30 seconds of downtime. Do I have your permission to proceed?"
- Developer: "Yes, approved."
- "Done. Instance upgraded. New CPU: 45%. PR created with the changes."

What Makes AEOS Different

Comparison Matrix

Feature	ChatGPT	Copilot	AWS DevOps	AEOS
Understands Code	✓	✓	✗	✓
Accesses Infrastructure	✗	✗	✓	✓
Autonomous Execution	✗	✗	✗	✓
Human Approval for Dangerous Actions	✗	✗	✗	✓
Remembers Past Fixes	✗	✗	✗	✓
Multi-Cloud Support	✗	✗	✗	✓
Ephemeral Workspaces	✗	✗	✗	✓
Plugin-Driven Extensibility	✗	✗	✗	✓

Real-World Examples

Example 1 — Fixing a Broken Build

Step	What Happens	Who Does It
1	Developer pushes broken code	Developer
2	CI/CD pipeline fails	GitHub Actions
3	User asks: "Fix the failing build in repo X"	User
4	AEOS clones the repo into an ephemeral workspace	Main Engine
5	AEOS reads the error logs	GitHub Plugin Agent
6	Agent identifies the missing import statement	GitHub Plugin Agent
7	Agent fixes the code and runs tests	GitHub Plugin Agent
8	Tests pass, pull request is created automatically	GitHub Plugin Agent
9	Workspace is destroyed	Main Engine
10	User reviews and merges the PR	User

Time saved: roughly 30 minutes down to 2 minutes.

Example 2 — Cloud Infrastructure Optimisation

Step	What Happens	Who Does It
1	User: "My EC2 instance is slow, upgrade it"	User
2	AEOS checks CloudWatch metrics	Main Engine → AWS Plugin
3	Agent sees 95% CPU utilisation	AWS Agent
4	Agent recommends t2.micro → t3.medium	AWS Agent
5	Paused — human approval required	Main Engine
6	User sees: "Downtime ~30 seconds. Approve?"	User
7	User approves	User
8	Agent stops instance, modifies type, restarts	AWS Agent
9	Agent verifies new CPU is 45%	AWS Agent
10	Task complete	AWS Agent

Time saved: roughly 15 minutes down to 3 minutes.

Example 3 — Multi-Cloud Debugging (Cross-Domain)

Step	What Happens	Who Does It
1	User: "My K8s pod can't connect to AWS RDS"	User
2	Main Engine splits the task into two sub-tasks	Main Engine
3	Task A: K8s plugin checks pod logs and network policies	K8s Agent
4	Task B: AWS plugin checks RDS security groups	AWS Agent
5	K8s Agent reports pod is timing out on port 5432	K8s Agent
6	AWS Agent reports the security group is missing the cluster CIDR	AWS Agent
7	AWS Agent adds the CIDR to the security group	AWS Agent
8	K8s Agent verifies the pod can now connect	K8s Agent
9	Both agents confirm the issue is resolved	Main Engine
10	User receives a unified report	User

Time saved: roughly 1 hour down to 5 minutes.

System Architecture

The "Lego Brick" Model

At the centre sits the AEOS Main Engine, acting as the project manager for every request. It routes tasks to the right specialist, manages security credentials, spins up temporary workspaces, and streams progress back to the user in real time.

AEOS MAIN ENGINE *"The Project Manager" — routes tasks, manages credentials, creates ephemeral workspaces, streams live status*		
AWS Plugin *"The Cloud Specialist"* • EC2 • RDS • S3 • CloudWatch	**K8s Plugin** *"The Container Specialist"* • Pods • Deployments • Services • Helm Charts	**GitHub Plugin** *"The Code Specialist"* • Read code • Fix bugs • Run tests • Create PRs

The Flow

- The user asks AEOS to do something, for example "Fix my broken build."
- The Main Engine understands what is needed.
- The Main Engine selects the right Plugin Agent, or several.
- The Plugin Agent reasons about the problem using AI.
- The Plugin Agent acts, using tools such as the AWS CLI, kubectl, or the GitHub API.
- If the action is dangerous, it pauses and asks for approval.
- If the action is safe, it executes and reports the result.
- The outcome is stored so AEOS can recall it next time.

Key Features

1. Plugin-Driven Architecture

> *The platform has zero capabilities by default. Plugins are installed to give it new abilities.*

- Install the AWS Plugin ⟶ AEOS can manage AWS
- Install the K8s Plugin ⟶ AEOS can manage Kubernetes
- Install the GitHub Plugin ⟶ AEOS can fix code and create pull requests

2. Ephemeral Workspaces (Security First)

> *Code is cloned, fixed, tested, and then deleted. Only the fix and the pull request remain.*

- No permanent storage of user code
- No risk of data leaks
- Every session starts fresh

3. Approval Gate (Human-in-the-Loop)

> *The AI never performs a dangerous action without explicit permission.*

- No automatic infrastructure deletion
- No automatic production changes
- The user sees exactly what is proposed, and why, before it happens

4. Knowledge Layer (Continuous Learning)

> *AEOS remembers every fix it has ever made.*

- If the same error recurs, the known fix is applied instantly
- If a similar incident occurs, AEOS suggests the past solution

Technology Stack

Component	Technology	Why
Main Engine	Python + FastAPI	Fast, modern, easy to build on
AI Framework	LangGraph + LangChain	Standard for building reliable AI agents
AI Models	OpenAI API, Claude API, Local LLMs	Flexible and cost-optimised
Plugin Runtime	Docker Containers	Isolated, secure, portable
Credential Vault	HashiCorp Vault	Industry-standard secret management

Component	Technology	Why
Vector Database	Qdrant / ChromaDB	Stores past fixes and knowledge
Frontend	React / Streamlit	Interactive, real-time UI

Live Demonstration Walkthrough

1. Starting point

A broken GitHub repository is shown, with the failing GitHub Action visible.

2. The request

AEOS is asked: "Fix the failing build in repo XYZ."

3. The live stream

```
[10:00:01]  Global Orchestrator: Received request
[10:00:02]  Classifying request... Detected: GitHub Actions
[10:00:05]  Launching GitHub Plugin Agent
[10:00:08]  GitHub Agent: Analysing error logs...
[10:00:10]  Tool Call: get_workflow_logs()
[10:00:12]  GitHub Agent: Found missing dependency
[10:00:15]  Tool Call: write_file(package.json)
[10:00:18]  Tool Call: run_command(npm test)
[10:00:22]  Tests passed
[10:00:25]  Tool Call: create_pr()
[10:00:28]  Pull Request #42 created
```

4. The pull request

The automatically created PR is opened and reviewed live in GitHub.

5. A cross-domain question

AEOS is asked: "My K8s pod can't connect to AWS RDS."

6. Agent collaboration

The interface shows the K8s Agent and AWS Agent working in parallel on their respective sub-tasks.

7. The approval gate

A prompt appears: "Human Approval Required." Once approved, the change is applied.

8. Knowledge recall

AEOS is asked: "Fix high CPU on EC2." It responds: "Similar incident found — applying the known fix."

Project Scope and Depth

Criteria	How AEOS Meets It
Technical Depth	LangGraph, LangChain, Docker, Kubernetes, AWS SDKs
System Design	Microservices, plugin architecture, event-driven design
AI Integration	Multi-agent systems, RAG, vector databases
Security	Credential vault, least-privilege access, approval gates
Scalability	Plugin-driven, containerised, cloud-native
Real-World Impact	Addresses genuine, everyday DevOps problems

Project Summary

> *AEOS is a plugin-driven autonomous engineering platform that acts like a virtual engineering team rather than a coding assistant. It fixes builds, manages cloud infrastructure, and writes code — all with human approval at every step.*

What It Does

- Fixes broken CI/CD pipelines
- Upgrades cloud infrastructure
- Patches security vulnerabilities
- Creates pull requests with validated fixes
- Remembers past incidents

What Makes It Safe

- Human approval required for dangerous actions
- Code is never permanently stored
- Credentials are never exposed to the AI
- Every action is audited

What Makes It Scalable

- Plugin-driven architecture
- New plugins can be added for AWS, Kubernetes, GitHub, databases, monitoring, security, and more
- Third-party developers can build their own plugins

Anticipated Questions

How is this different from ChatGPT?

ChatGPT can only talk about infrastructure — it doesn't know a user's AWS setup and gives suggestions. AEOS connects directly to the cloud with the user's credentials and executes validated fixes with approval.

What if the AI makes a mistake?

Three safety layers guard against this: ephemeral workspaces test changes in isolation first, an approval gate requires explicit permission for anything dangerous, and a maximum retry limit stops the agent and asks for help after repeated failures.

Can it delete a production database?

No. Any tool capable of deleting a database is marked as requiring approval. The interface blocks execution and displays a clear "Human Approval Required" prompt — nothing happens until the user explicitly approves it.

What if someone adds a malicious plugin?

Every plugin must declare its permissions in a manifest, which is shown to the user before installation. Plugins also run inside isolated Docker containers and cannot access the main system directly.

How are costs managed?

AEOS uses tiered AI models — inexpensive models handle simple tasks like classification, while more capable models are reserved for complex reasoning about infrastructure. This approach cuts API costs by roughly 70%.

Closing Thought

> *Most AI projects build a better chatbot. AEOS builds a better engineering team. It is not about answering questions — it is about getting work done.*